

CPE 494 / CPE 643

Applied Machine Learning

Course Syllabus for the 1st Semester of the year 2026

Department of Computer Engineering, KMUTT

Thursday 18:00PM – 20:00PM, Room CPE1114

FACEBOOK Group: **CPE-AML-2026** <https://www.facebook.com/groups/27935903616045220>

Instructor

Asst. Prof. Dr. Jumpol Polvichai

Tel. 02-470-9388

email: jumpol@gmail.com

Course Description

Hands-on experiences in implementation of several advance topics in machine learning field. Many learning techniques are covered such as inductive learning, naïve Bayesian, reinforcement learning, genetic programming, artificial neural networks, and stochastic neural networks, in a simulation robotic application.

Course Learning Outcomes

1. Understand about several new topics of machine learning techniques
2. Implement simple ML applications related to each topic in a simulation robot problem.

Course Outline

August 6	First Meeting: SimBot & Simple Reactive Robot Programming	Assignment 1
August 13	Fuzzy Logic Control	Assignment 2
August 20	Implementing Fuzzy Logic Control in SimBot	
August 27	Genetic Algorithms	Assignment 3
September 3	Implementing Genetic Algorithms in SimBot	
September 10	Artificial Life	Assignment 4
September 17	Implementing Artificial Life in SimBot	
October 1	Introduction to Neural Networks	Assignment 5
October 8	Implementing Neural Networks in SimBot	
October 29	Decision Making	Assignment 6
November 5	Reinforcement Learning (Q-Learning)	
November 12	Implementing Q-Learning in SimBot	
November 19	Naïve Bayesian Classification	Assignment 7
November 26	Implementing Naïve Bayesian Classification in SimBot	
December 11	Final Exam	

Evaluations

Final 30%, Assignment Coding 35% and Assignment Presentation 35%

Textbooks

- Stuart Russell and Peter Norvig, **Artificial Intelligence: A Modern Approach**, Prentice Hall, 2003.
- Tom Mitchell, **Machine Learning**, McGraw Hill, 1997.

August 2026

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
						1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29
30	31					

September 2026

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
		1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30			

October 2026

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	31

November 2026

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30					

December 2026

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
		1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30	31		